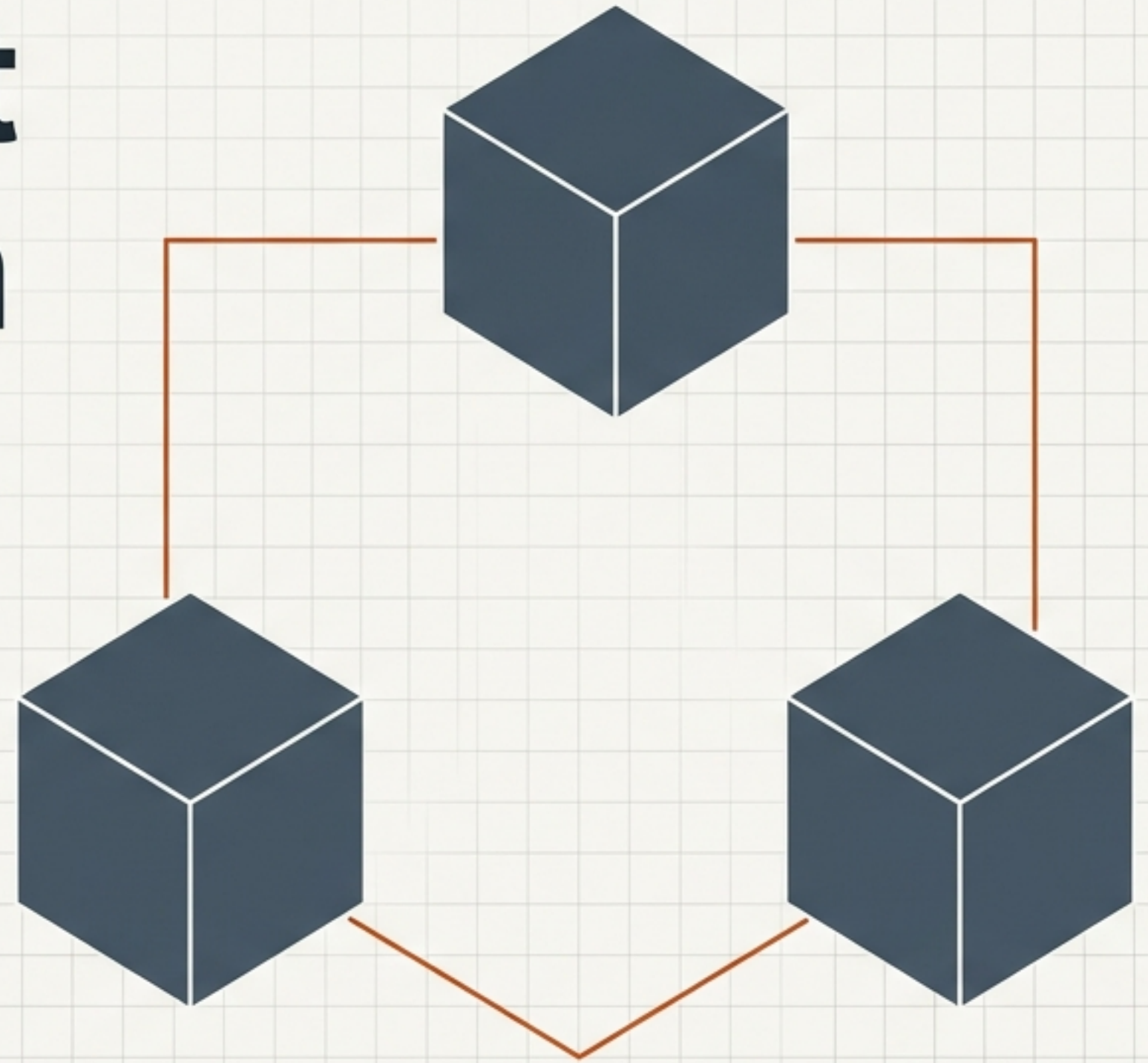
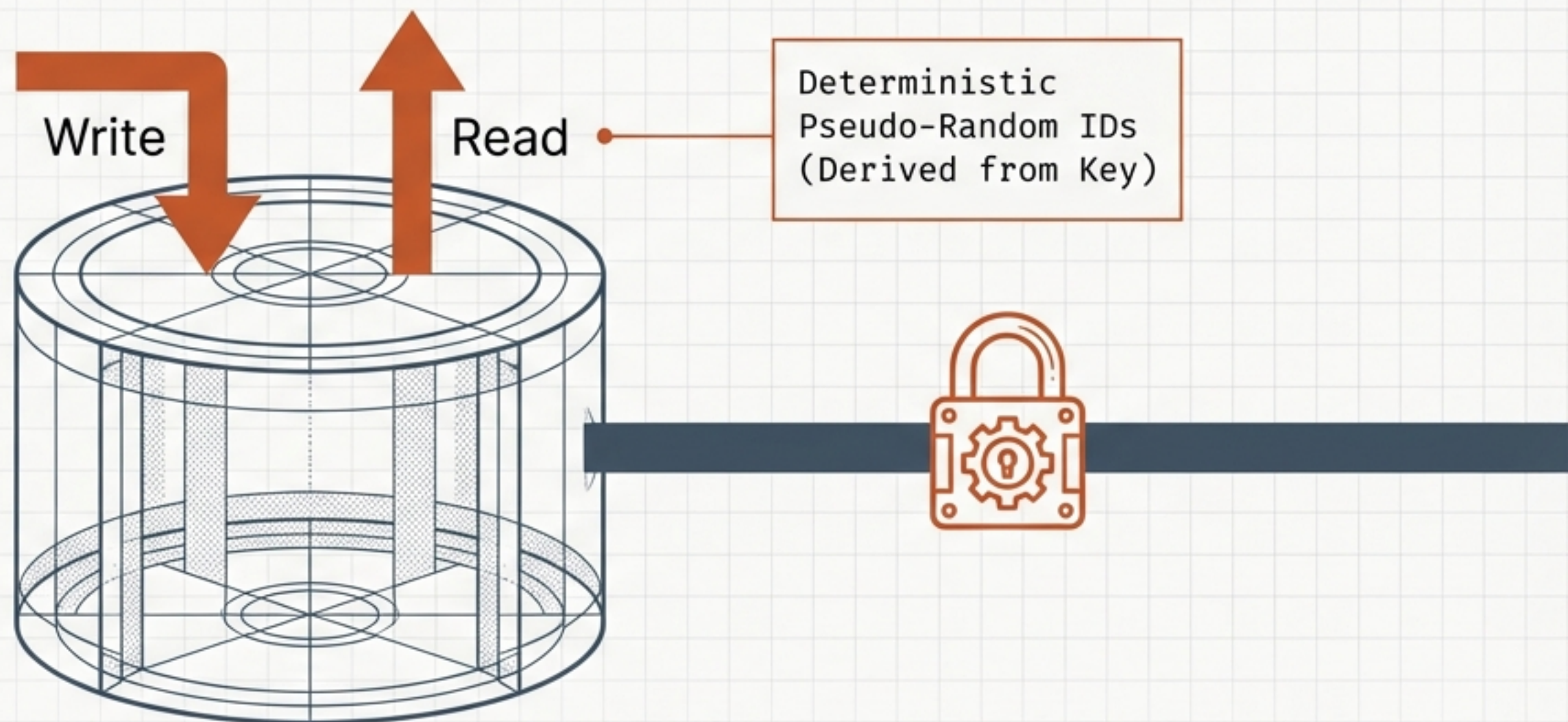


Agent-to-Agent Communication

Vaults as a Signed Message Plane



High-performance vault I/O unlocks a new communication plane.



Agents can now read, write, and branch on a vault without cloning the entirety of it.

By eliminating the full-clone bottleneck, Vaults transition from static, heavy storage into a dynamic, real-time message routing layer.

Current agent workflows lack fundamental cryptographic guarantees.



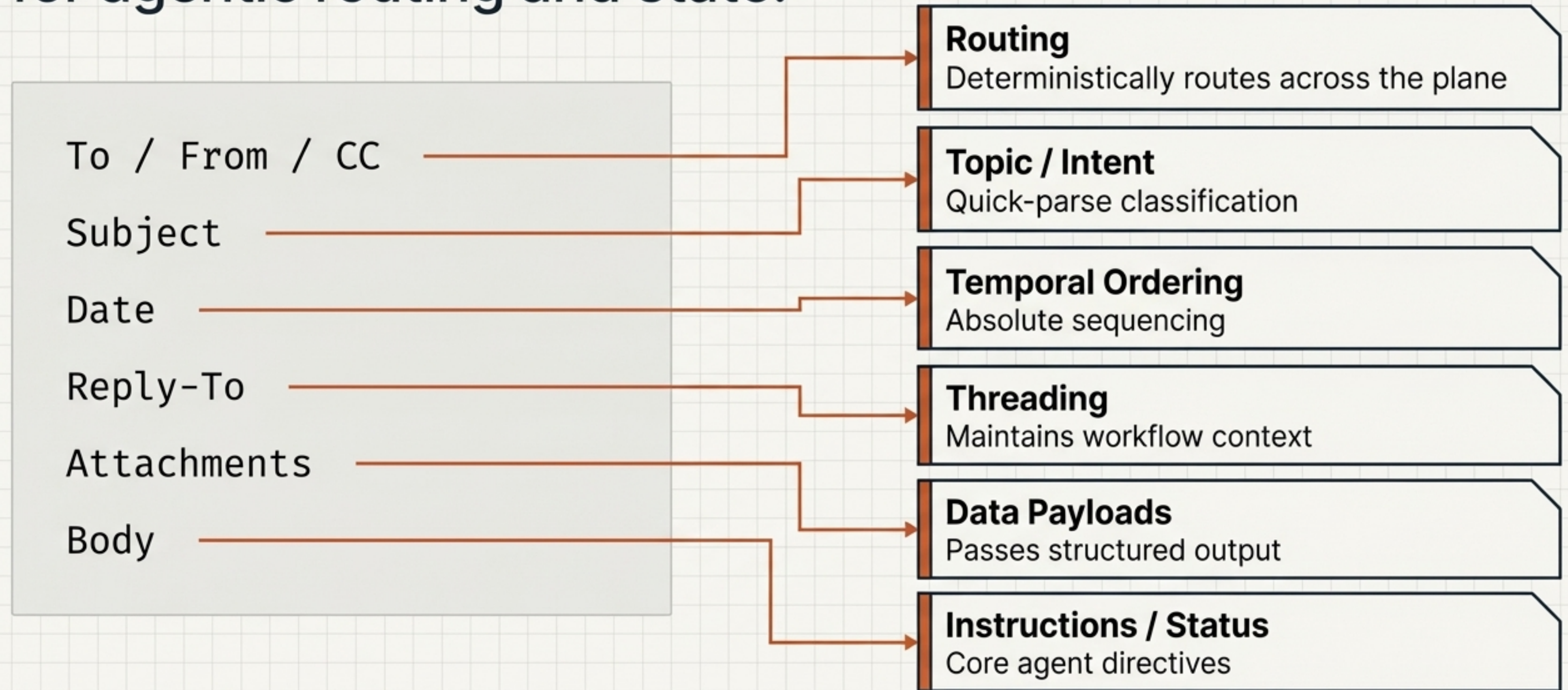
Capability Gap	Architectural Risk
Authentication	Cannot definitively prove Agent B's message actually originated from Agent A.
Confidentiality	Cannot mathematically guarantee only the intended agent reads a payload.
Sequencing	Cannot strictly enforce execution order (Agent B waiting for Agent A).
Provenance	Cannot trace the exact, verifiable history of who, what, and when.
Replay & Debug	Cannot reliably rewind, inspect, and reproduce complex agent workflows.
Prompt Defence	Cannot prevent an agent from reading system instructions meant for a different agent.

A signed message plane built entirely from existing primitives.



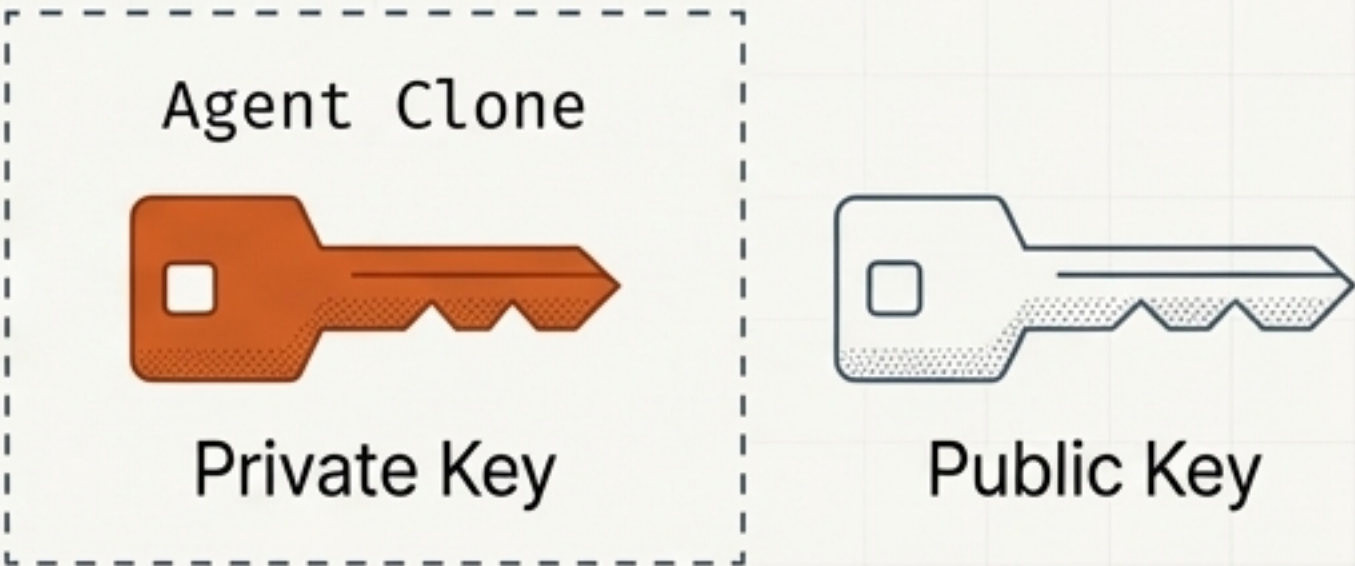
We resolve all six architectural deficits by reusing the exact cryptographic primitives already designed and battle-tested for human workflows.

Repurposing the EML standard for agentic routing and state.



PKI per agent translates mathematical properties into architectural trust.

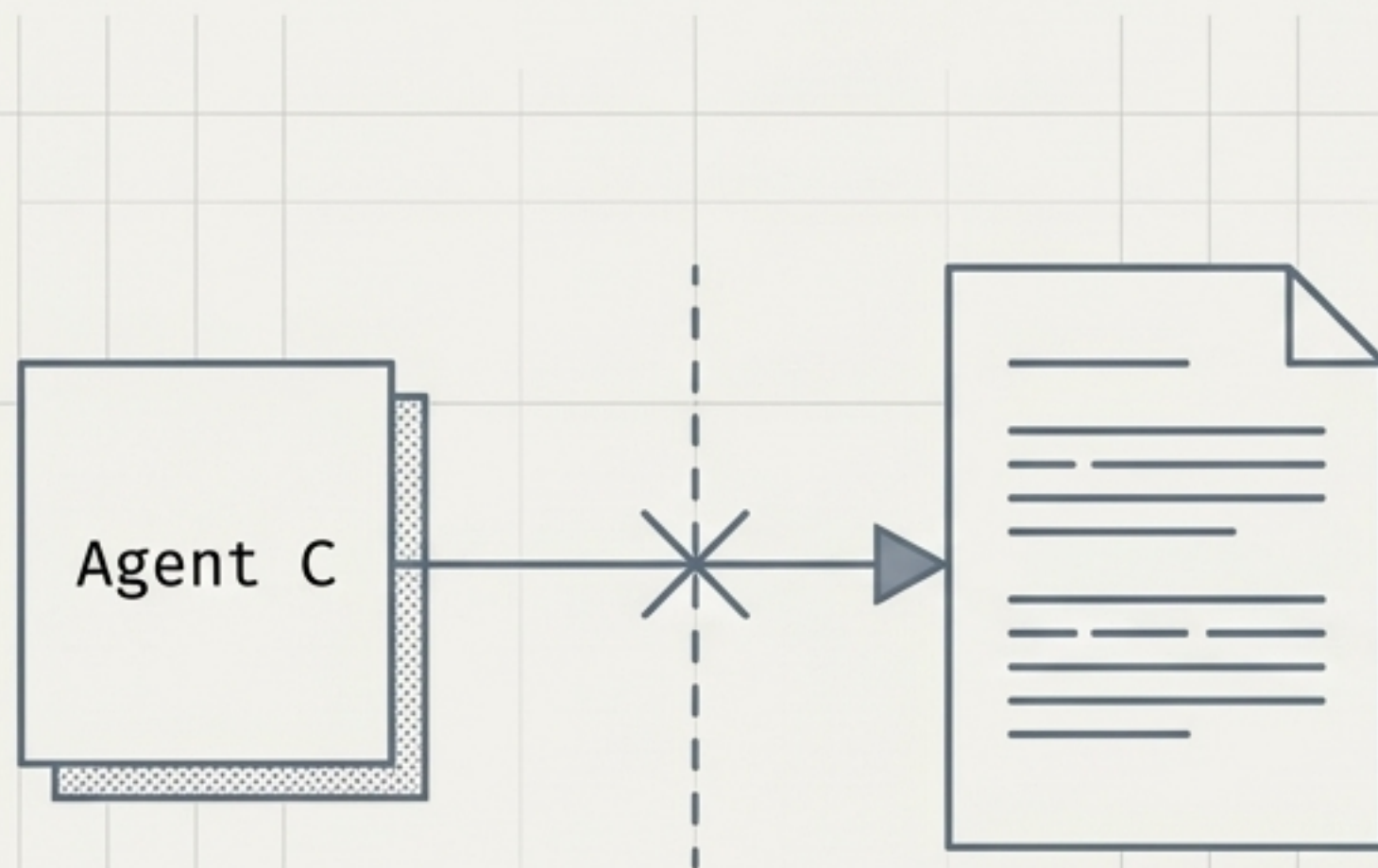
Inherits the proven branch model pattern: every clone automatically generates a unique key pair.



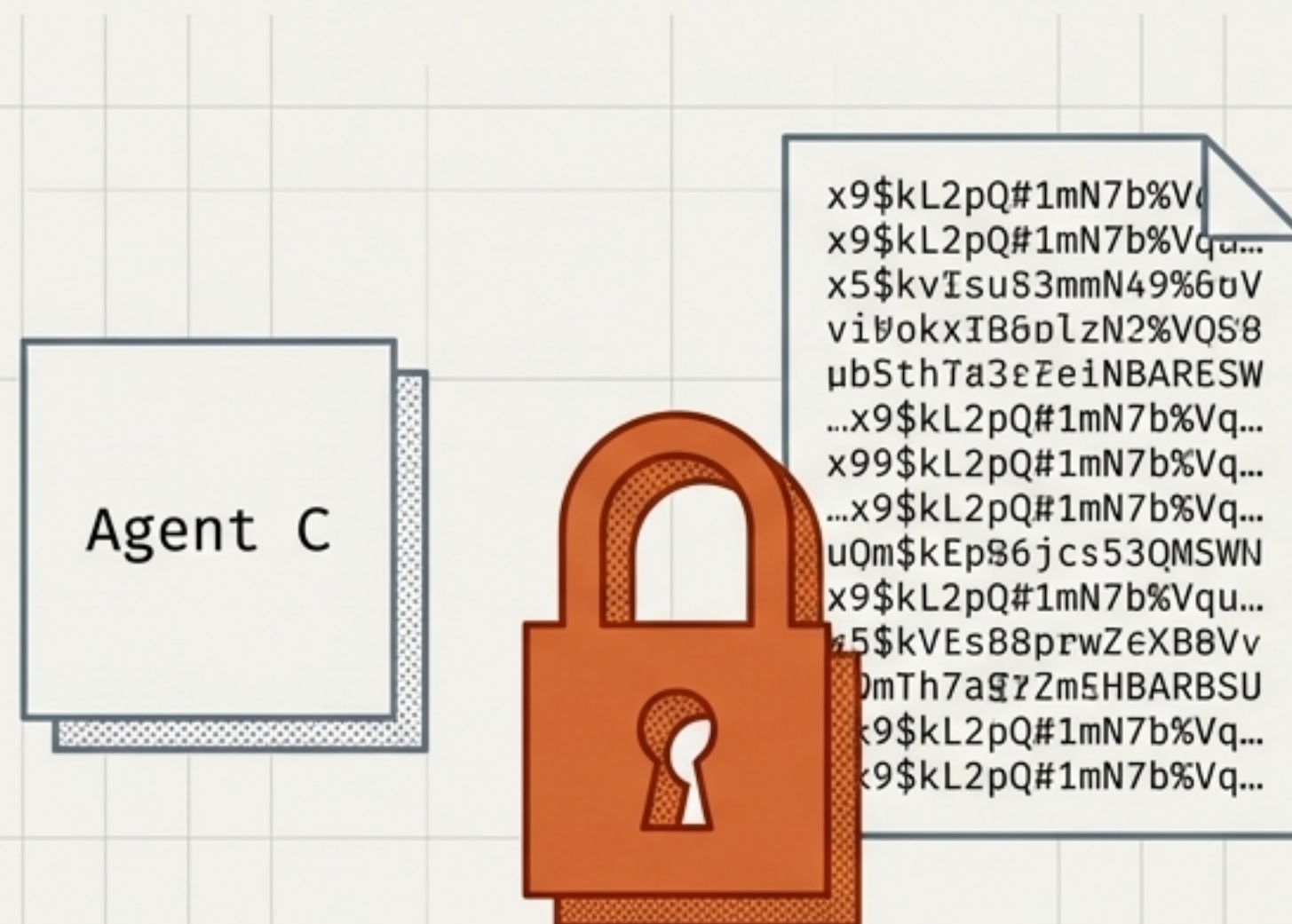
Mathematical Property	Architectural Guarantee
Signed Messages	Validates origin. Agent B mathematically verifies the message is from Agent A (prevents hallucinated or injected workflows).
Encrypted Messages	Validates destination. Only the holder of Agent B's private key can decrypt the contents.

Prompt injection defence transitions from policy to physics.

If Agent C attempts to read Agent B's instructions, it fails **entirely**. This failure is not governed by a bypassable access control list. It is governed by **cryptography**. The “cannot vs. will not” principle applied to agent security.

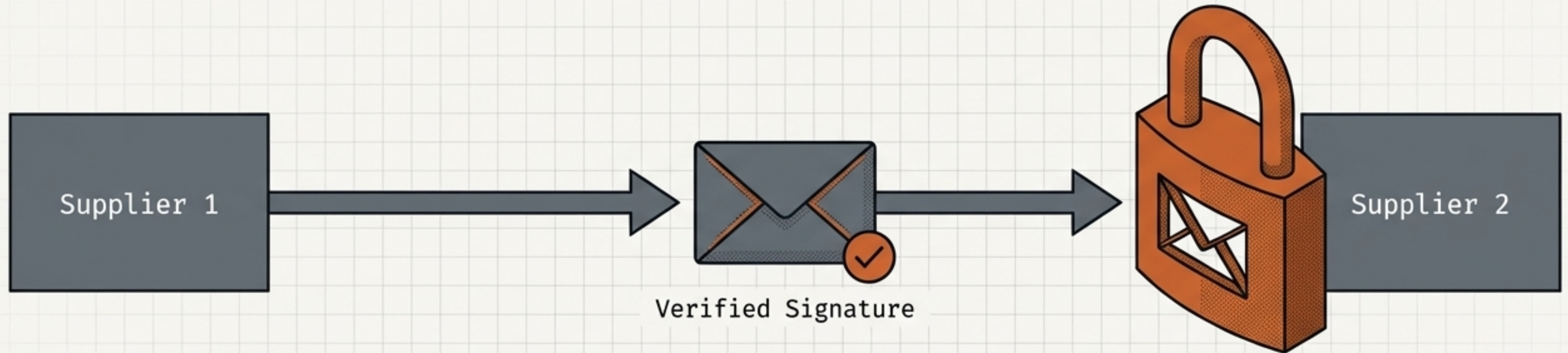


Will Not
(Software Logic)



Cannot
(Mathematical Physics)

Sequential workflows enforced by encryption chains, not logic.



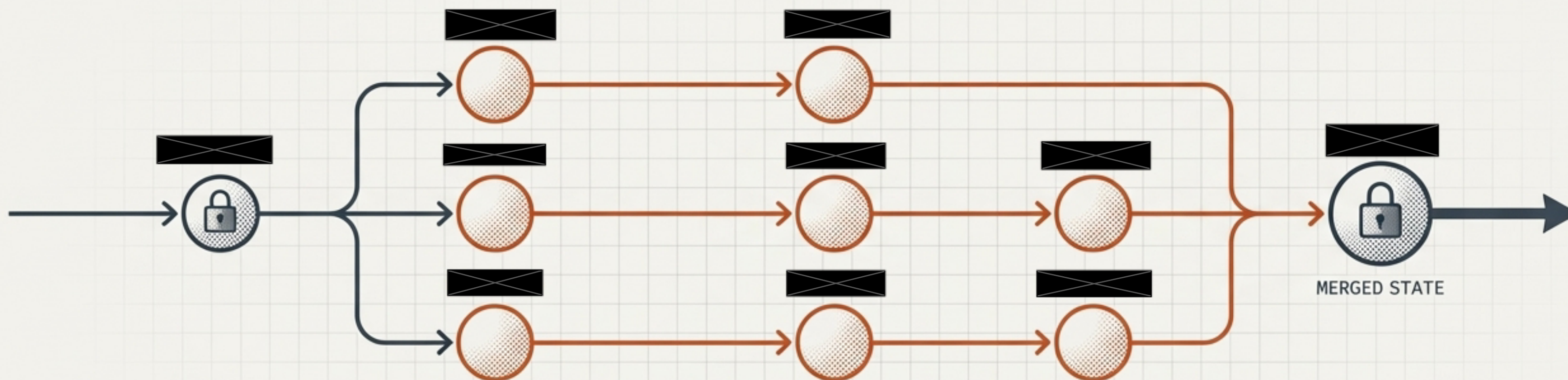
Mechanic

Supplier 2 cannot execute prematurely because the trigger message mathematically does not exist yet.

Security

Supplier 2 cannot forge the trigger message to bypass the wait state because it lacks Supplier 1's private key. Sequence becomes an unbreakable cryptographic chain.

Vault branching natively handles parallel agent execution.



Mechanic

When workflows diverge, each parallel agent operates strictly within its own isolated vault branch.

Resolution

Branches merge only when parallel tasks conclude. Conflicts are isolated and resolved natively at merge time, entirely eliminating runtime collisions during live execution.

Immutable vault history enables absolute state control.



T-01



T-02



T-03



T-04



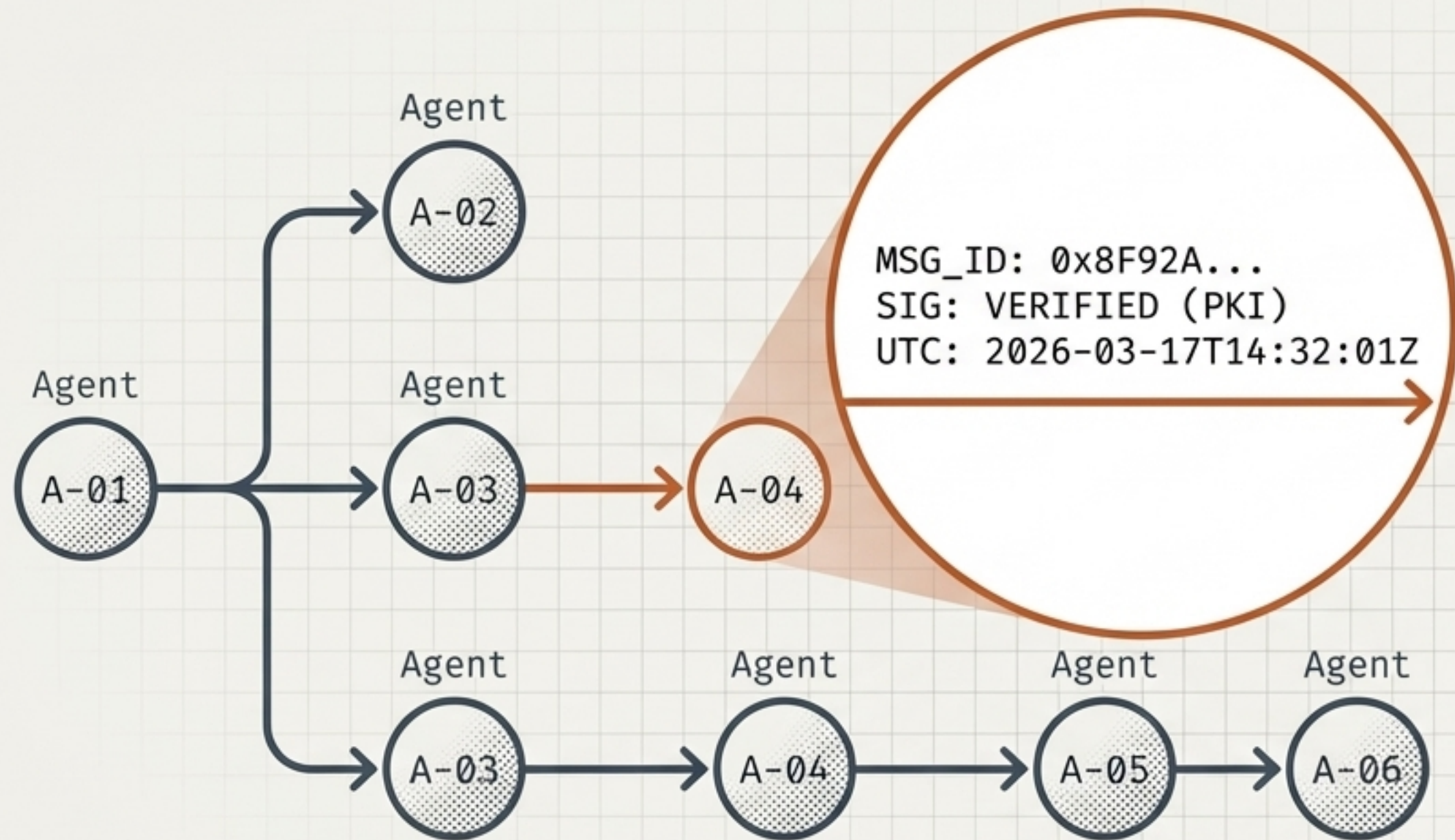
T-05



T-06

Start / Stop	Replay	Fork
Pause active workflows instantly by halting message delivery. Resume by releasing the queue.	Clone the vault at any historical time T. Replay the exact signed sequence to perfectly reproduce agent behaviour.	Branch the vault from any historical timestamp to test alternative approaches or prompts without corrupting the main execution.

The execution graph provides an ultimate, verifiable audit trail.



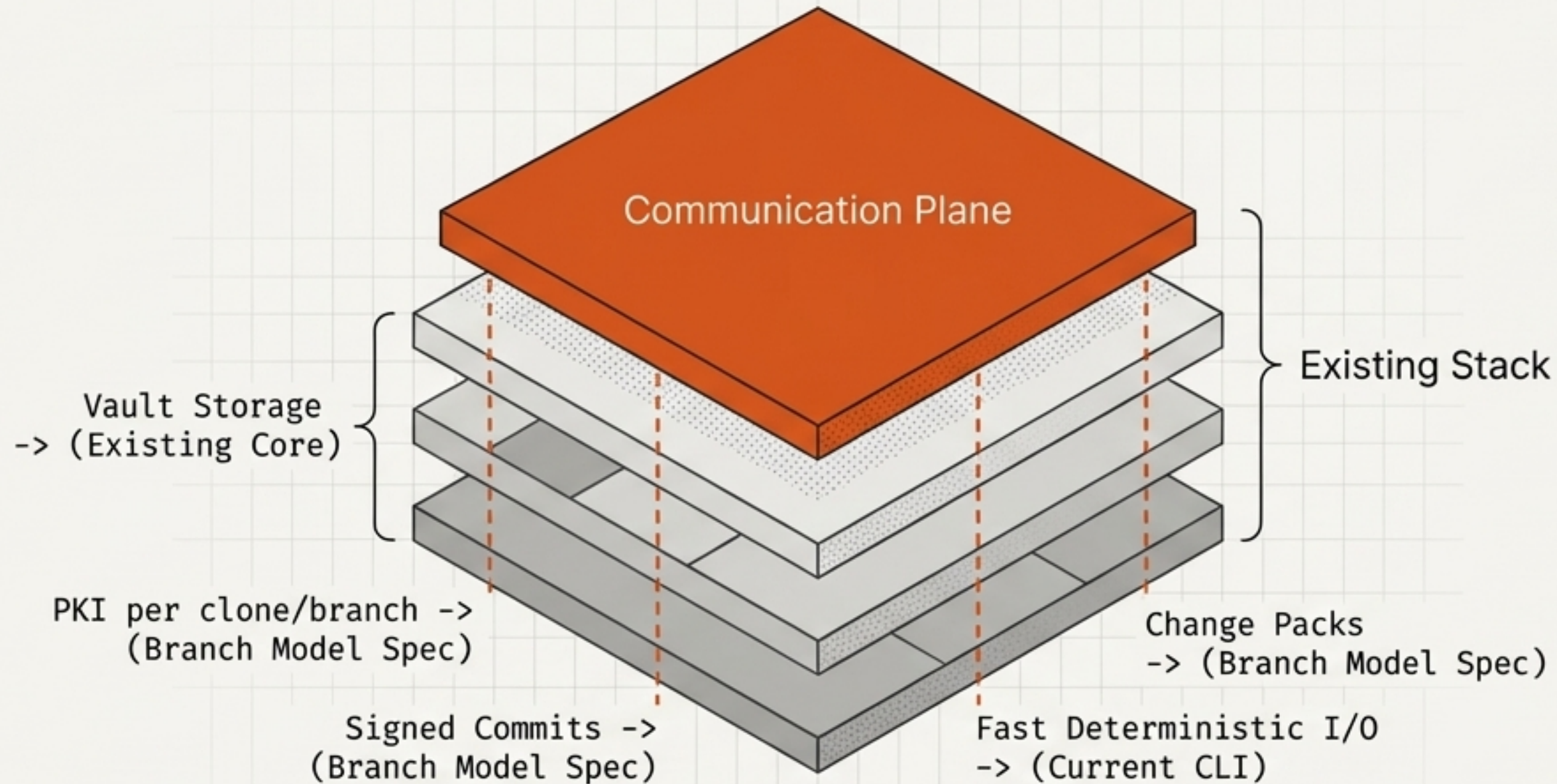
Insight

The entire execution flow natively resolves into a graph of signed messages.

Impact

Serves as a flawless debugging interface for engineers and an irrefutable explainability tool for AppSec and governance. Trace any autonomous decision to its exact cryptographic source.

Architectural synthesis requires zero new underlying primitives.



We introduce only a format convention (EML) and a routing pattern (encrypt-for-recipient).
The required foundation is already built.

v0.16.17 Acceptance Criteria

- [] EML-based message format defined for agent-to-agent communication.
- [] Agent key pair generation implemented (isolated private keys).
- [] Outbound message signing active.
- [] Inbound message encryption active.
- [] Sequential enforcement demoed (Agent B mathematically blocked until Agent A signs).
- [] Prompt injection defence demoed (Agent fails to read unencrypted ciphertext).
- [] Message graph visualisation active (Who, What, Whom, When).